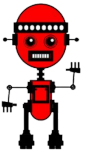




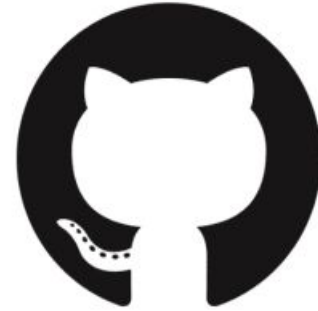
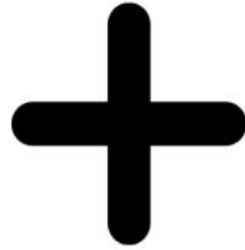
visit us : <https://learnotronix.robotronix.co.in/>



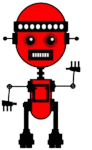
Git & GitHub



Git



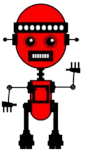
GitHub



CONTENT

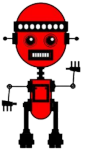


- What is Git and GitHub?
- Why it is so important?
- What is version?
- Important Commands of Git



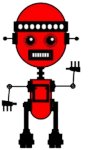
What is Git ?

- Git is a distributed version control system (VCS) that allows developers to track changes in their code, collaborate with others, and manage project history.
- It was created by Linus Torvalds and is designed to be fast and efficient.
- Git operates on your local machine, which means you can work on your code even without an internet connection.
- It tracks changes in files and allows you to commit those changes with a meaningful message.
- You can create branches to work on different features or bug fixes independently.



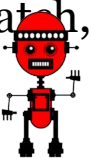
What is GitHub ?

- GitHub is a web-based platform that uses Git as its underlying version control system.
- It provides a cloud-based hosting service for Git repositories.
- GitHub offers collaboration features such as issue tracking, pull requests, code reviews, and project management tools.
- Developers can use GitHub to store their code repositories and collaborate with other team members or the open-source community.



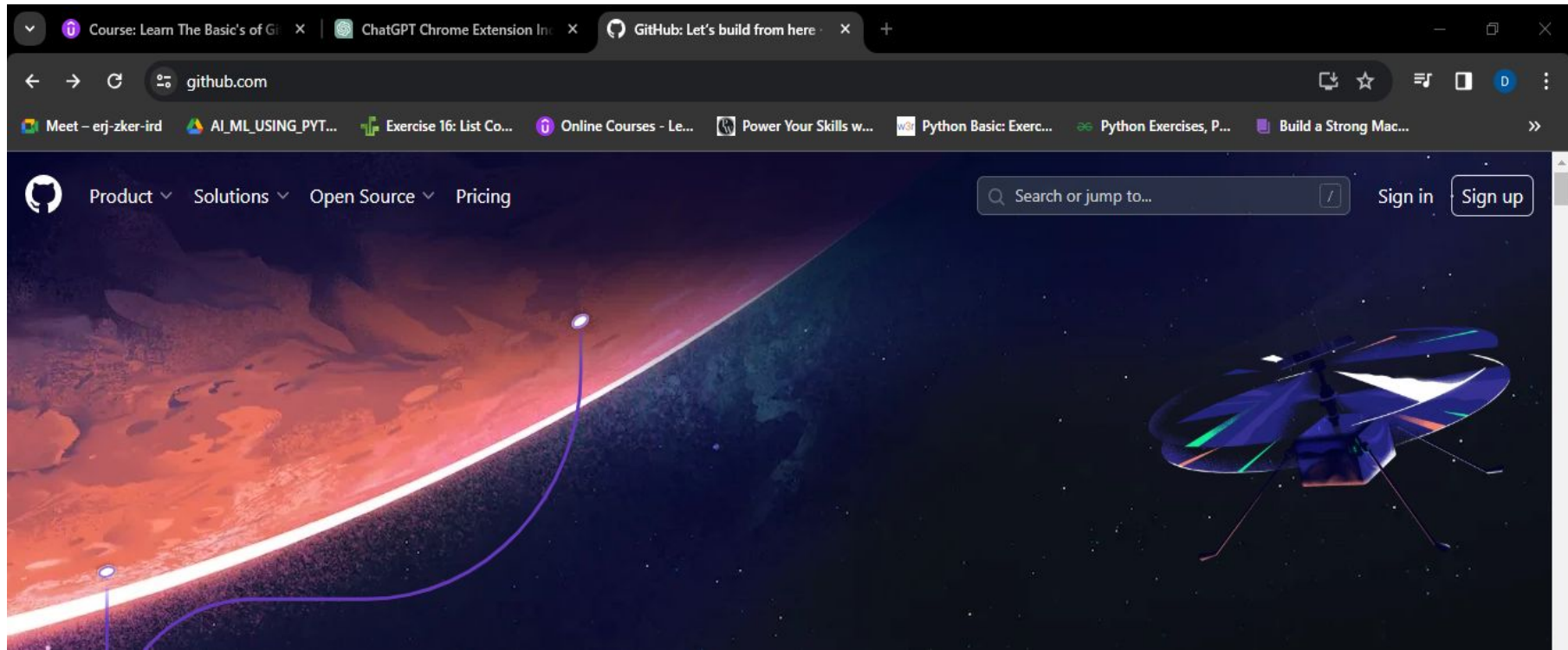
What is Version ?

- In the context of Git, a "version" typically refers to a specific state or snapshot of a codebase at a particular point in time. Git is a distributed version control system, and it tracks the history and changes made to a codebase by creating a series of commits. Each commit represents a version of the codebase, and these versions are identified by a unique hash (SHA-1) that represents the contents and history of that specific commit.
- In the context of software, a "version" refers to a specific release or iteration of a software application or product. Software versions are typically identified using a version number or code, and they help developers, users, and other stakeholders track the evolution and state of a software product.
- Versions are often represented using numbers or codes, such as "1.0," "2.0.1," or "v3.0-beta." These numbers can have different components (major, minor, patch, etc.) to convey the significance of changes between versions.



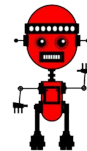
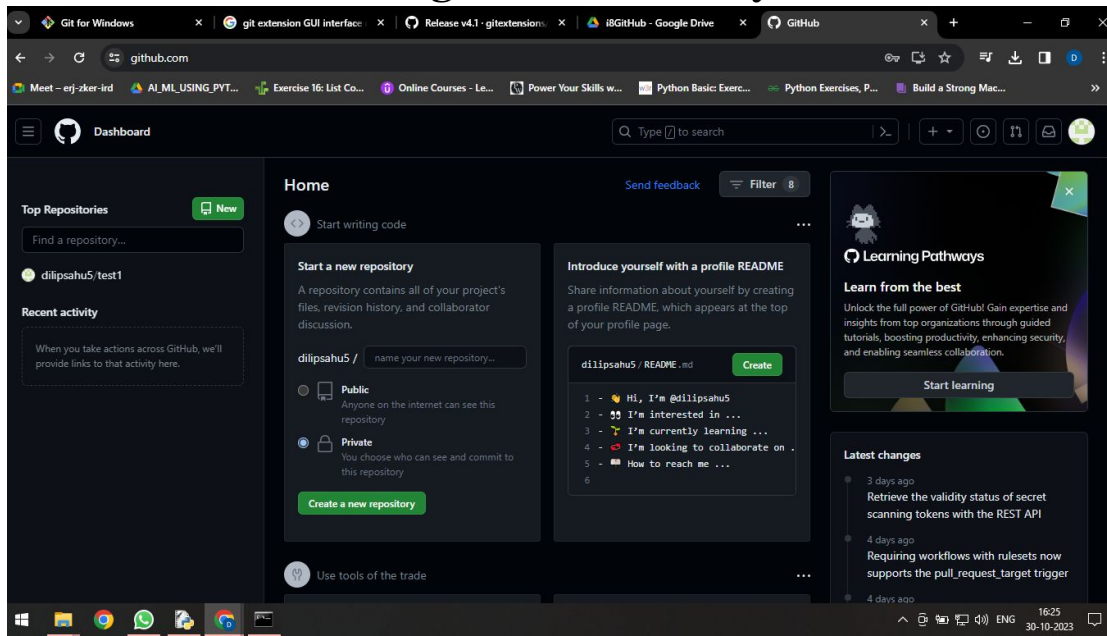
Steps to Create Account on Github.com

1. Open the <https://github.com> in your browser
2. Click on on Signup

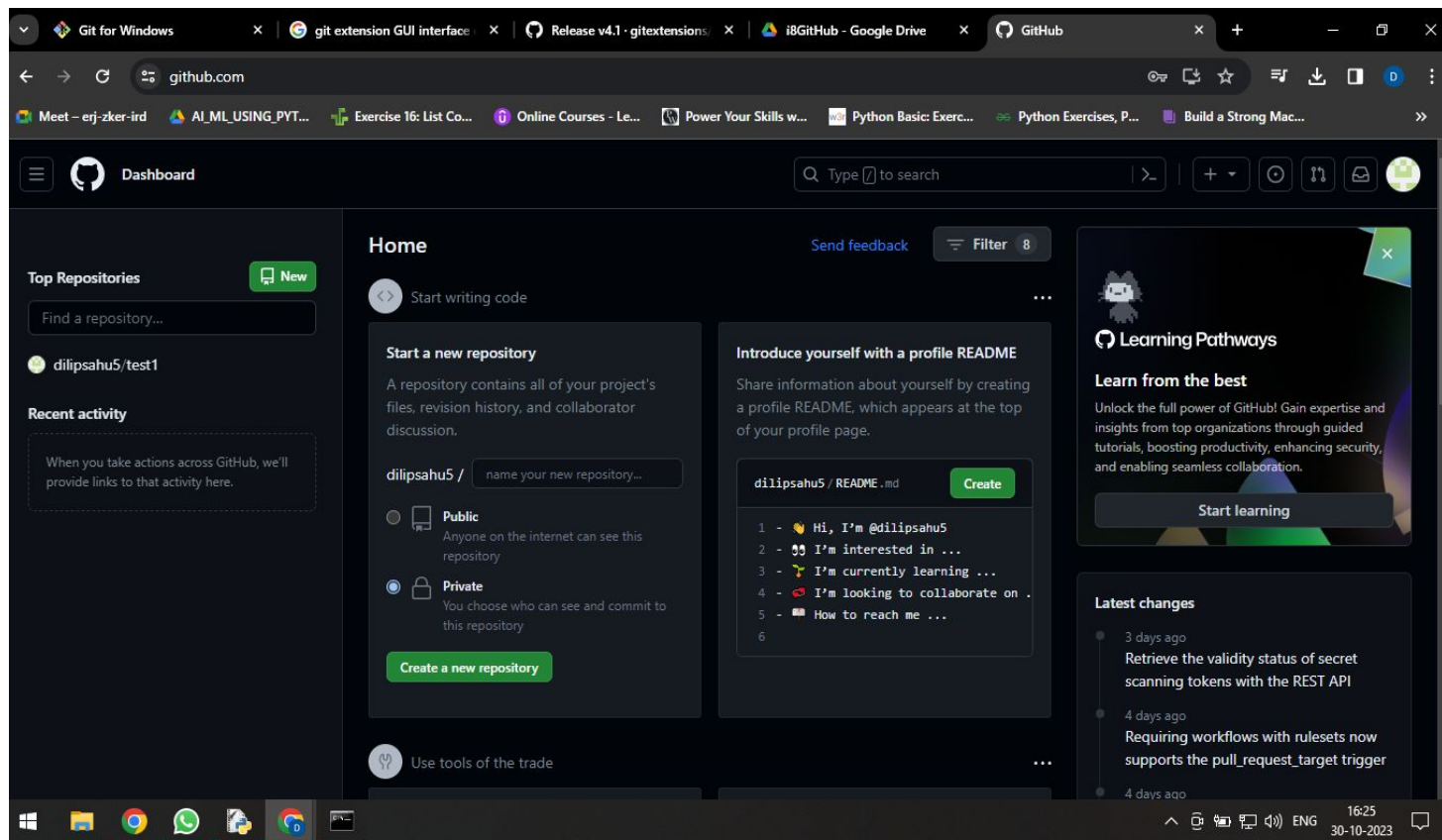


Steps to Create Account on Github.com

1. Open the <https://github.com> in your browser
2. Click on on Signup
3. Fill the all needful information and follow the instruction given by github.com
4. Login in github account with login credentials you reach at this page



Steps to Create Account on Github.com



The screenshot shows the GitHub.com homepage in a web browser. The browser's address bar displays 'github.com'. The page features a dark theme with a sidebar on the left containing 'Top Repositories' (listing 'dilipsahu5/test1') and 'Recent activity'. The main content area is titled 'Home' and includes a 'Start writing code' button. Below this, there are two primary sections: 'Start a new repository' and 'Introduce yourself with a profile README'. The 'Start a new repository' section offers options for 'Public' (visible to anyone) and 'Private' (visible only to collaborators), with a 'Create a new repository' button. The 'Introduce yourself with a profile README' section shows a preview of a README file for 'dilipsahu5 / README.md' with a 'Create' button. To the right, there are promotional cards for 'Learning Pathways' and 'Latest changes'. The 'Learning Pathways' card encourages users to 'Learn from the best' by unlocking the full power of GitHub through guided tutorials. The 'Latest changes' card lists recent updates, such as retrieving the validity status of secret scanning tokens and supporting workflows with rule sets. The bottom of the page shows a Windows taskbar with various application icons and a system clock indicating 16:25 on 30-10-2023.

Git for Windows x git extension GUI interface x Release v4.1 · gitextensions x iGitHub - Google Drive x GitHub x

github.com

Meet - erj-zker-ird AI ML USING PYT... Exercise 16: List Co... Online Courses - Le... Power Your Skills w... Python Basic: Exerc... Python Exercises, P... Build a Strong Mac...

Dashboard

Type to search

Home

Send feedback Filter 8

Start writing code

Start a new repository

A repository contains all of your project's files, revision history, and collaborator discussion.

dilipsahu5 / name your new repository...

☐ Public
Anyone on the internet can see this repository

☒ Private
You choose who can see and commit to this repository

Create a new repository

Introduce yourself with a profile README

Share information about yourself by creating a profile README, which appears at the top of your profile page.

dilipsahu5 / README.md Create

```
1 - Hi, I'm @dilipsahu5
2 - I'm interested in ...
3 - I'm currently learning ...
4 - I'm looking to collaborate on .
5 - How to reach me ...
6
```

Learning Pathways

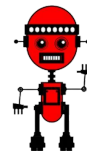
Learn from the best

Unlock the full power of GitHub! Gain expertise and insights from top organizations through guided tutorials, boosting productivity, enhancing security, and enabling seamless collaboration.

Start learning

Latest changes

- 3 days ago
Retrieve the validity status of secret scanning tokens with the REST API
- 4 days ago
Requiring workflows with rule sets now supports the pull_request_target trigger
- 4 days ago

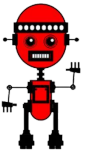


Install the git for windows

1. Open the link for command line git <https://gitforwindows.org/>
2. Install the downloaded software by click next next on button
3. After the completion of installation open the command prompt from windows and run the command as below
> git --version

```
C:\Users\DILIP>git --version  
git version 2.42.0.windows.2  
  
C:\Users\DILIP>
```

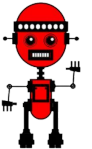
You will get the version of git if installation was proper



Cheat Sheet of Git Commands

<https://ndpsoftware.com/git-cheatsheet.html#loc=stash;>

<https://github.com/joshnh/Git-Commands/blob/master/README.md>



Git Commands

use - Initialize the git folder on local system

git init

Syntax : git init

We have to navigate to our project directory and type the command git init to initialize a Git repository for our local project folder. Git will create a hidden .git directory and use it for keeping its files organized in other subdirectories.

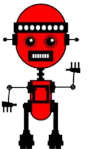
```
Command Prompt
Microsoft Windows [Version 10.0.19045.3570]
(c) Microsoft Corporation. All rights reserved.

C:\Users\DILIP>D:

D:\>cd git_test

D:\git_test>git init
Initialized empty Git repository in D:/git_test/.git/

D:\git_test>
```



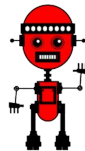
Add the remote repo to local repo

Use - add remote repo with local repo via http link

Syntax : git remote add origin <https://github.com/dilipsahu5/test1.git>

```
D:\git_test>git remote add origin https://github.com/dilipsahu5/test1.git  
D:\git_test>
```

If no error or any message appear means we have successfully added the remote repo to local repo.



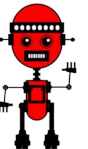
To check already a remote repo is added to local repo or not

> git remote -v

```
D:\git_test>git remote -v
origin https://github.com/dilipsahu5/test1.git (fetch)
origin https://github.com/dilipsahu5/test1.git (push)

D:\git_test>
```

It shows the list of remote repo connected with local repo



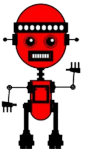
If you want to remove the remote repo from the local repo

> git remote remove origin

```
D:\git_test>git remote remove origin  
D:\git_test>
```

To confirm run

```
D:\git_test>git remote -v  
D:\git_test>
```

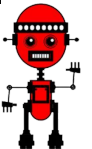


To add again we run same command

```
D:\git_test>git remote add origin https://github.com/dilipsahu5/test1.git  
D:\git_test>
```

To confirm run

```
D:\git_test>git remote -v  
origin https://github.com/dilipsahu5/test1.git (fetch)  
origin https://github.com/dilipsahu5/test1.git (push)  
D:\git_test>
```



add any file to local folder to check the status

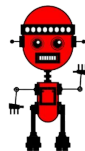
```
D:\git_test>git status
On branch master

No commits yet

Untracked files:
  (use "git add <file>..." to include in what will be committed)
    test_file.txt

nothing added to commit but untracked files present (use "git add" to track)
D:\git_test>
```

It shows list of files you have created and not tracked

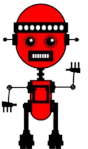


To stage a file to make ready to commit

```
D:\git_test>git add test_file.txt  
D:\git_test>
```

Check the status again and look changed color

```
D:\git_test>git status  
On branch master  
  
No commits yet  
  
Changes to be committed:  
  (use "git rm --cached <file>..." to unstage)  
    new file:   test_file.txt  
  
D:\git_test>
```

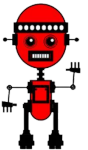


To stage all file to make ready to commit

```
D:\git_test>git add --all  
D:\git_test>
```

To stage all file to make ready to commit

```
D:\git_test>git add .  
D:\git_test>
```

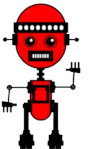


To unstage a file to remove from the commit

```
D:\git_test>git rm --cached test_file.txt  
rm 'test_file.txt'  
  
D:\git_test>
```

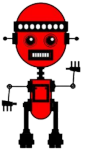
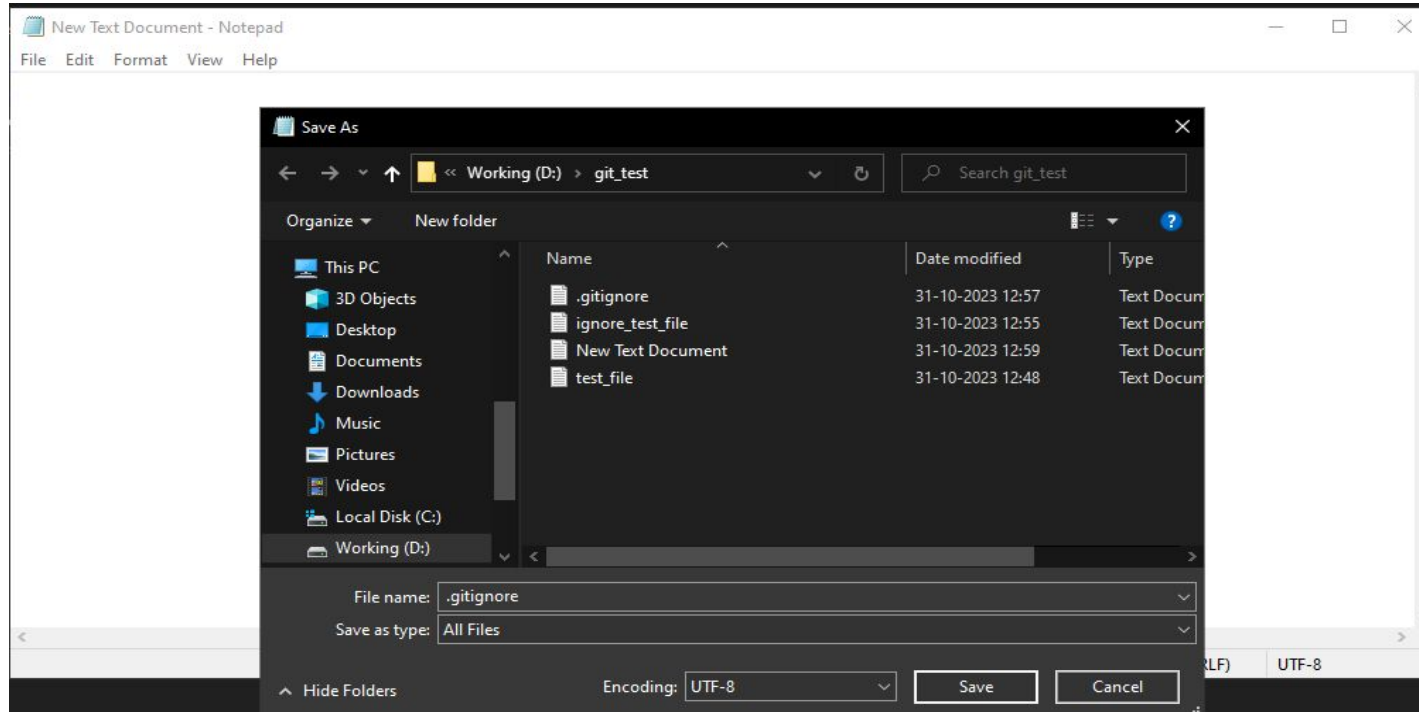
To check status of file

```
D:\git_test>git status  
On branch master  
  
No commits yet  
  
Untracked files:  
  (use "git add <file>..." to include in what will be committed)  
    test_file.txt  
  
nothing added to commit but untracked files present (use "git add" to track)
```



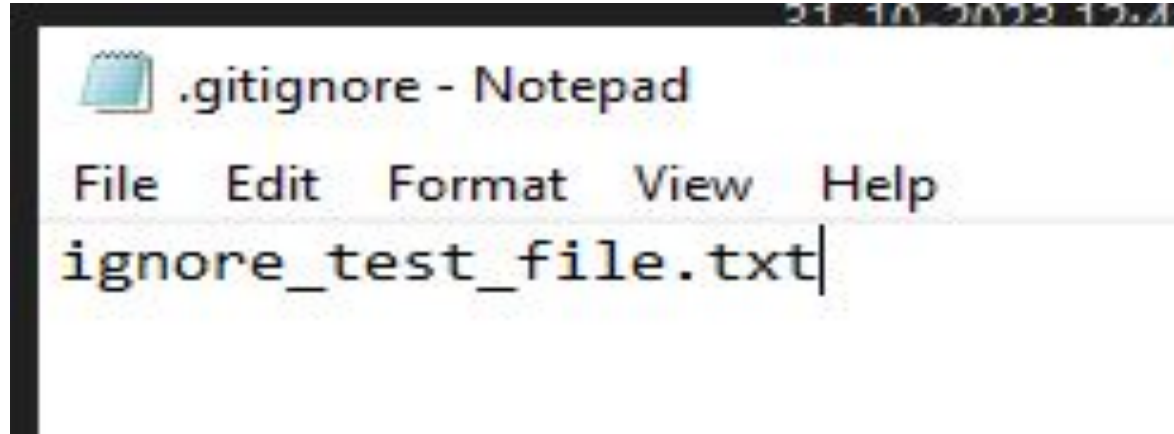
To ignore file from git tracking

1. Open a notepad text file and save it with name **.gitignore** and file type all type in save option

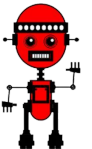


To ignore file from git tracking

2. Open that file and write the name of file with extension which you want to ignore from git track and save the file



3. Check the status of git. The file name you have written in the .gitignore will not show here



To ignore file from git tracking

```
D:\git_test>git status
On branch master

No commits yet

Changes to be committed:
  (use "git rm --cached <file>..." to unstage)
        new file:   test_file.txt

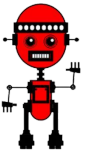
Untracked files:
  (use "git add <file>..." to include in what will be committed)
        .gitignore
        ignore_test_file.txt

D:\git_test>git status
On branch master

No commits yet

Changes to be committed:
  (use "git rm --cached <file>..." to unstage)
        new file:   test_file.txt

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        .gitignore
```



To configure the global user with username and email



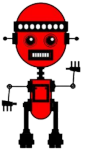
```
D:\git_test>git config --global user.name "dilipsahu5"
D:\git_test>git config --global user.email "tally.dilipsahu@gmail.com"
D:\git_test>
```

Emails

tally.dilipsahu@gmail.com – Primary
This email will be used for account-related notifications

> git config --global user.name "dilipsahu5"

> git config --global user.email "tally.dilipsahu@gmail.com"



To commit the file

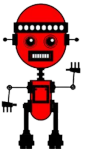
Before commit we must sure that we have stage the file (git add .file) and it status files are green in color

```
D:\git_test>git commit -m "intial commit"
[master (root-commit) 84c8045] intial commit
 2 files changed, 1 insertion(+)
 create mode 100644 .gitignore
 create mode 100644 Test_file_git.txt.txt

D:\git_test>
```

> git commit -m "Message for commit"

This is not a push. (Files are not uploaded on github)



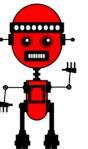
To Check the commits log which are commits done before

> git log

```
D:\git_test>git log
commit 84c804567f0ba180c3f6676a9d7817bb1bc96d49 (HEAD -> master)
Author: dilipsahu5 <tally.dilipsahu@gmail.com>
Date:   Fri Nov 3 14:02:11 2023 +0530

    intial commit

D:\git_test>
```



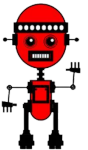
To push the files on github repo

> git push <name of connection when add the remote repo>

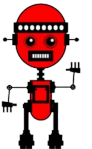
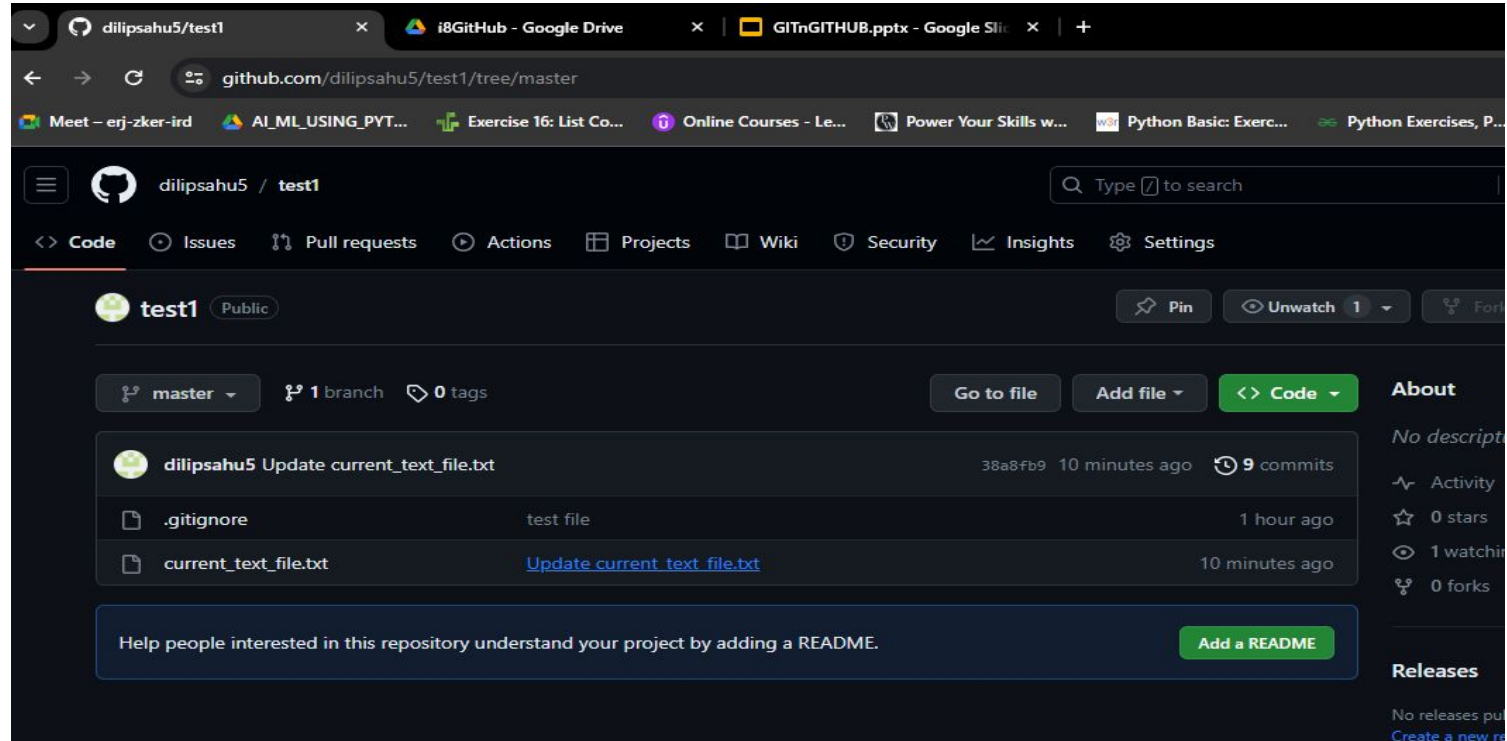
```
D:\git_test>git push origin master
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 4 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (4/4), 286 bytes | 286.00 KiB/s, done.
Total 4 (delta 0), reused 0 (delta 0), pack-reused 0
To https://github.com/dilipsahu5/test1.git
 * [new branch]      master -> master

D:\git_test>
```

On running this command a prompt will open in web browser and ask for you github email and password to authenticate the commitor . Provide the credential files will pushed to github

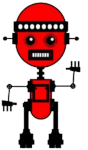


Now open your github account and check your repo the all committed files will be found here. It means you have successfully pushed all file.



Now if you want to push some files again you have to pull the files from the github and then repeat the whole process from staging to till push

- 1. `git pull origin master` (git command)**
- 2. make change in your file present in local repo and stage the changed file**
- 3. `git add .` (git command)**
- 4. `git commit -m "message what you have changed"` (git command)**
- 5. `git push origin master` (git command)**



THANKS



visit us : <https://learnotronix.robotronix.co.in/>

